

Davis M. Welakuh

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Summary

- A highly skilled High-Performance Computing (HPC) Application Engineer with focus on computational physics and chemistry software. Extensive experience in developing, optimizing, and scaling scientific and engineering applications for large-scale HPC & AI systems.
- Proficient in parallel programming, performance analysis, and system architecture, with expertise in leveraging multi-core processors, GPUs, and distributed computing frameworks. Adept at working with industry-standard HPC tools and libraries to enhance performance and efficiency across a variety of domains including computational physics, chemistry, and molecular dynamics.
- Theoretical and computational quantum physicist with 9 years expertise in quantum chemistry, quantum optics, condensed matter physics, nonlinear optics and quantum entanglement. Scientific communicator with at least 13 publications in scientific journals, more than a dozen talks at national and international conferences. Practical problem-motivated scientific software programmer in FORTRAN, Python, MATLAB, and IBM's Qiskit and for large-scale simulations/calculations. Active developer of an open-source computational physics/chemistry software Octopus.

Work Experience

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| Feb. 2024 – Present | ■ Senior HPC Application Engineer: Computational Physics & Chemistry
Department: HPC Benchmarking
Hewlett Packard Enterprise, New York NY, USA. <ul style="list-style-type: none">– Develop and optimize HPC applications for computational physics/chemistry simulations on multi-core CPU and GPU-based systems, improving performance over 30%.– Collaborate with scientific researchers to profile and identify bottlenecks in large-scale computational physics/chemistry applications, optimizing memory usage and reducing computational times.– Perform system-level tuning, benchmarking, and performance evaluations on national supercomputing clusters such as Frontier and NERSC. |
| Jul. 2023 – Jan. 2024 | ■ Postdoctoral Research Fellow.
Department: Simons Center for Computational Physical Chemistry.
New York University, New York NY, USA. |
| Jul. 2022 – Jun. 2023 | ■ Postdoctoral Research Scholar.
Department: Chemistry and Biochemistry.
University of California, Los Angeles CA, USA. |
| Jun. 2021 – Jun. 2023 | ■ Postdoctoral Research Scholar.
Department: John A. Paulson School Of Engineering And Applied Sciences.
Harvard University, Cambridge MA, USA. |
| Sep. 2016 – May 2021 | ■ Doctoral Research Fellow (IMPRS-UFAST).
Department: Theory department.
Max Planck Institute for the Structure and Dynamics of Matter, Hamburg, Germany. |

Education

- Sep. 2016 – Jun. 2021  Universität Hamburg, Hamburg, Germany.
Ph.D. in Theoretical Physics.
Thesis title: *Ab initio Strong Light-Matter Theoretical Framework for Phenomena in Non-relativistic Quantum Electrodynamics.*
- Oct. 2013 – Jul. 2015  University of Buea, South-West Region, Cameroon.
M.Sc. in Physics (First Class honors).
Thesis title: *Trapping, storage and retrieval of single-photon solitons in the quantum states of collective atomic excitations.*
- Oct. 2010 – Jul. 2013  University of Buea, South-West Region, Cameroon.
B.Sc. in Physics with minor in Computer Sciences.
Thesis title: *Solitons and Modulation instability in Optical fibers.*

Skills

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| Programming/HPC |  | C/C++, Fortran, Python, Matlab, Mathematica, OpenMP, OpenMPI, CUDA, HIP, Bash, Vi/Vim, SLURM/PBS/SGE. |
| Compilers |  | GNU, Intel/Intel oneAPI, AOCC, NVCC, Cray. |
| Development Tools |  | CMake/Make, Git, EasyBuild, Spack |
| Numerical Libraries |  | BLAS, LAPACK, ScaLAPACK, FFTW, NETCDF, HDF5 and many more. |
| HPC benchmark applications |  | NAMD, GROMACS, LAMMPS, AMBER, NWCHEM, CP2K, VASP, Quantum Espresso, Gaussian, Octopus. |
| Operating Systems |  | Linux, Unix, Windows OS, macOS. |
| Markup Language |  | LaTeX, Markdown. |
| Visualization & Design |  | Tikz-LATEX, gnuplot, Matplotlib, gimp, Inkscape, EdrawMax. |
| Languages |  | English (native), French and German. |

Scholarships and Fellowships

- Jul. 2023 – Jan. 2024  **Simons Postdoctoral Fellowship for Computational Physical Chemistry.**
New York University, NY USA.
- Sep. 2016 – Jun. 2021  **Doctoral Fellow at the International Max Planck Research School for Ultrafast Imaging & Structural Dynamics (IMPRS-UFAST).**
Max Planck Institute for the Structure and Dynamics of Matter, Hamburg, Germany.
- Oct. 2011 – Jul. 2016  **Presidential Grant of Excellence.**
University of Buea, South-West Region, Cameroon.
- Jul. 2015  **Awards:** Best graduating M.Sc. in Physics student (first-class honors) at the University of Buea, Cameroon (cash prize award).

Research Publications

Journal Articles

-  **Davis M. Welakuh**, V. Rokaj, M. Ruggenthaler, and A. Rubio, “Nonperturbative mass renormalization effects in nonrelativistic quantum electrodynamics,” *Phys. Rev. Research*, vol. 7, no. 1, p. 013 093, Jan. 2025.  DOI: 10.1103/PhysRevResearch.7.013093.

- 2 F. Tay, S. Sanders, A. Baydin, *et al.*, “TeraHertz chiral photonic-crystal cavities for Dirac gap engineering in graphene,” *Nature Communications*, vol. 16, pp. 2041–1723, 1 Jun. 2025.  DOI: 10.1038/s41467-025-60335-x.
- 3 **Davis M. Welakuh** and P. Narang, “Nonlinear optical processes in centrosymmetric systems by cavity-induced symmetry breaking,” *ACS Photonics*, vol. 11, pp. 369–377, 2 Jan. 2024.  DOI: 10.1021/acsp Photonics.2c01933.
- 4 **Davis M. Welakuh**, S. Tserkis, S. E. Smart, and P. Narang, “Cavity-mediated molecular entanglement and generation of non-classical states of light,” *J. Phys. Chem. A*, vol. 128, pp. 799–806, 4 Jan. 2024.  DOI: 10.1021/acs.jpca.3c05779.
- 5 S. E. Smart, **Davis M. Welakuh**, and P. Narang, “Many-body excited states with a contracted quantum eigensolver,” *J. Chem. Theory Comput.*, vol. 20, no. 9, pp. 3580–3589, 2024.  DOI: 10.1021/acs.jctc.4c00030.
- 6 **Davis M. Welakuh** and P. Narang, “Tunable and efficient harmonic generation from strongly coupled light-matter system,” *ACS Photonics*, vol. 10, pp. 383–393, 2 Jan. 2023.  DOI: 10.1021/acsp Photonics.2c00966.
- 7 **Davis M. Welakuh**, J. Flick, M. Ruggenthaler, H. Appel, and A. Rubio, “Frequency-dependent Sternheimer linear-response formalism for strongly coupled light-matter systems,” *J. Chem. Theory Comput.*, vol. 18, pp. 4354–4365, 7 Jun. 2022.  DOI: 10.1021/acs.jctc.2c00076.
- 8 **Davis M. Welakuh** and P. Narang, “Transition from Lorentz to Fano spectral line shapes in non-relativistic quantum electrodynamics,” *ACS Photonics*, vol. 9, pp. 2946–2955, 9 Aug. 2022.  DOI: 10.1021/acsp Photonics.2c00256.
- 9 **Davis M. Welakuh**, M. Ruggenthaler, M.-L. M. Tchenkoue, H. Appel, and A. Rubio, “Down-conversion processes in ab-initio non-relativistic quantum electrodynamics,” *Phys. Rev. Research*, vol. 3, no. 3, p. 033067, Jul. 2021.  DOI: 10.1103/PhysRevResearch.3.033067.
- 10 J. Flick, **Davis M. Welakuh**, M. Ruggenthaler, H. Appel, and A. Rubio, “Light-matter response in nonrelativistic quantum electrodynamics,” *ACS Photonics*, vol. 6, no. 11, pp. 2757–2778, Oct. 2019.  DOI: 10.1021/acsp Photonics.9b00768.
- 11 V. Rokaj, **Davis M. Welakuh**, M. Ruggenthaler, and A. Rubio, “Light-matter interaction in the long-wavelength limit: No ground-state without dipole self-energy,” *Journal of Physics B: Atomic, Molecular and Optical Physics*, vol. 51, no. 3, p. 034005, 2018.  DOI: 10.1088/1361-6455/aa9c99.
- 12 **Davis M. Welakuh** and A. M. Dikandé, “Storage and retrieval of time-entangled soliton trains in a three-level atom system coupled to an optical cavity,” *Optics Communications*, pp. 27–33, Nov. 2017.  DOI: 10.1016/j.optcom.2017.06.082.
- 13 M. L. D. Tchenkoue, **Davis M. Welakuh**, and A. M. Dikandé, “Soliton trains induced by adaptive shaping with periodic traps in four-level ultracold atom systems,” *Communications in Theoretical Physics*, vol. 67, no. 6, p. 667, Jun. 2017.  DOI: 10.1088/0253-6102/67/6/667.

Oral and Poster Presentations

Bibliography

- 1 **Davis M. Welakuh**, July 2017, Trieste, Italy.  URL: <https://codata.org/initiatives/data-skills/research-data-science-summer-schools/2017-research-data-schools/2017-research-data-science-applied-workshops/>.
- 2 **Davis M. Welakuh** and A. M. Dikandé, *Storage and retrieval of time-entangled soliton trains in a three-level atom system coupled to an optical cavity*, July 2016, Kigali, Rwanda.  URL: <https://home.infn.it/en/media-outreach/news-all/archive-news-press-release/77-news-infn->

it-archivio/1990-l-infn-all-african-school-of-physics-2016-per-rilanciare-la-ricerca-nel-continente-nero-2.

- 3 **Davis M. Welakuh**, J. Flick, M. Ruggenthaler, H. Appel, and A. Rubio, *Correlated electron-photon excitations from first-principles*, May 2019, Madrid, Spain (Oral).  URL: <https://dep.ftmc.uam.es/conference-announcements/xxv-international-summer-school-nicolas-cabrera-2018/>.
- 4 **Davis M. Welakuh**, J. Flick, M. Ruggenthaler, H. Appel, and A. Rubio, *Correlated electron-photon excitations from first-principles*, September 2018, Madrid, Spain (Poster).  URL: <https://dep.ftmc.uam.es/conference-announcements/xxv-international-summer-school-nicolas-cabrera-2018/>.
- 5 **Davis M. Welakuh**, J. Flick, M. Ruggenthaler, H. Appel, and A. Rubio, *Correlated electron-photon excitations within quantum electrodynamical density-functional theory*, March 2018, Los Angeles, California (Oral).  URL: <https://meetings.aps.org/Meeting/MAR18/Session/L03.9>.
- 6 **Davis M. Welakuh**, J. Flick, M. Ruggenthaler, H. Appel, and A. Rubio, *Correlated electron-photon excitations within quantum electrodynamical density-functional theory*, April 2018, Eindhoven, Netherlands (Poster).  URL: <https://surfacephotonics.org/scom-2018/#page-content>.
- 7 **Davis M. Welakuh** and P. Narang, *Nonlinear optical processes in centrosymmetric systems by cavity-induced symmetry breaking*, March 2022, Cambridge, Massachusetts (Oral-invited).
- 8 **Davis M. Welakuh** and P. Narang, *Transition from lorentz to fano spectral line shapes in non-relativistic quantum electrodynamics*, March 2022, Chicago, Illinois (Oral).  URL: <https://meetings.aps.org/Meeting/MAR22/Session/K46.7>.
- 9 **Davis M. Welakuh** and P. Narang, *Tunable and efficient harmonic generation from strongly coupled light-matter system*, November 2021, Boston, Massachusetts (Oral).
- 10 **Davis M. Welakuh**, V. Rokaj, M. Ruggenthaler, and A. Rubio, *Non-perturbative mass renormalization effects in non-relativistic quantum electrodynamics*, March 2024, Minneapolis, Minnesota (Oral).
- 11 **Davis M. Welakuh**, M. Ruggenthaler, M.-L. M. Tchenkoue, H. Appel, and A. Rubio, *Cavity controlled inverse harmonic generation*, March 2020, Denver, Colorado (Oral).  URL: <https://meetings.aps.org/Meeting/MAR20/Session/J05.9>.
- 12 **Davis M. Welakuh**, M. Ruggenthaler, M.-L. M. Tchenkoue, H. Appel, and A. Rubio, *High-harmonic generation with quantized fields: Minimal coupling of pauli spinors to quantized electromagnetic fields*, March 2019, Boston, Massachusetts (Oral).  URL: <https://meetings.aps.org/Meeting/MAR19/Session/C20.9>.
- 13 **Davis M. Welakuh**, S. Tserkis, S. E. Smart, and P. Narang, *Entangling molecules in the strong coupling regime*, March 2023, Las Vegas, Nevada (Oral).

Teaching & Mentorship

Teaching assistant

Oct. 2014 – Jun. 2016

- Responsibilities included developing new class materials, leading class discussions, supervising computational works, grading all assignments, and meeting with students individually.
 - Course Title: PHY 308: Quantum mechanics I. Institution: University of Buea. (40-50 students).
 - Course Title: PHY 410: Quantum mechanics II. Institution: University of Buea. (40-50 students).
 - Course Title: PHY 306: Mathematical methods of physics I. Institution: University of Buea. (40-50 students).

Mentorship experience

Oct. 2015 – Jul. 2016

- Mentored two undergraduate students and mentored one graduate student in the physics department at the University of Buea.

Referee Activities

Scientific journals

- Nature Communications, PLOS One.

References

Available on Request